

SUMMARY TABLE: MATHEMATICS GRADE 10

Sub-Strand 2.1: Similarity and Enlargement — Teacher Reference (Pre-filled)

SUMMARY TABLE: MATHEMATICS GRADE 10	
Sub-Strand	Sub-Strand 2.1: Similarity and Enlargement
Driving Question	DRIVING QUESTION: How can a passport photo be made bigger or smaller without changing its shape, and how do its length, area and cost change when we do so?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Same Photo, Two Sizes: Launching the Driving Question Board	We saw two prints of Achieng's passport photo: stamp size S (3 cm x 4 cm, KES 30) and postcard size P (6 cm x 8 cm, KES 120). The picture has the same shape on both prints, but the width, height, face height, diagonal and price are all different.	A picture can be made bigger without changing its shape, and we call such pictures similar. The original is the object and the bigger copy is the image. Different measurements change in different ways, and the price did not simply double.	We think the lengths on P are bigger than on S because the photo was enlarged, and the price is more than double because more paper is used, but we do not yet know the rule. We have started the DQB with questions about length, area and cost to investigate in later lessons.
Lesson 2 Spotting Similar Objects Around Us	We observed that photos S (3 x 4 cm) and P (6 x 8 cm) look the same shape but differ in size, while the distorted 6 x 4 photo looks stretched. We also saw that a flag and a smaller flag, a map and a smaller map, and a real matatu (700 cm) and its 1:25 model (28 cm) look the same shape.	We learned to sort objects as similar (same shape, size may differ) or not similar (shape changed, for example stretched). A model matatu is similar to a real matatu, and 'congruent' is used only when shape and size are both the same.	Achieng's prints S and P are similar because one is an enlargement of the other without stretching, while doubling only the width gives a distorted, non-similar photo. We still need numbers to test similarity and to explain how length, area and cost change.
Lesson 3 Testing Similarity: Equal Angles and Proportional Sides	We measured the angles and compared sides of triangles 3-4-5 and 6-8-10 (all ratios 2), rectangles 3 x 4 and 6 x 7 (ratios 2 and 1.75), and photo S (3 x 4 cm) against the distorted 6 x 4 cm photo (ratios 2 and 1). All the rectangles and photos had right angles.	Figures are similar only if corresponding angles are equal AND corresponding sides are in the same ratio. Equal angles alone are not enough, and one matching ratio is not enough.	Print P (6 x 8 cm) is similar to print S (3 x 4 cm) because all angles are 90 degrees and $6/3 = 8/4 = 2$. The photo made 6 x 4 cm is stretched because the width ratio is 2 but the height ratio is 1, so it is not similar. This helps explain how Achieng's photo can change size without changing shape.
Lesson 4 Finding the Linear Scale	We measured width, height, diagonal and face height of Achieng's prints S (3 x 4 cm),	The linear scale factor $k = \text{image length} / \text{object length}$, using the same units and	Print P is an enlargement of print S with LSF 2, so every length doubles by

Factor: How Many Times Bigger Is Achieng's Photo?	P (6 x 8 cm) and L (9 x 12 cm). For S to P, every length was 2 times bigger ($6/3 = 8/4 = 10/5 = 4.8/2.4 = 2$), and for S to L it was 3 times. In the distorted 6 x 4 photo, the width ratio was 2 but the height ratio was 1.	corresponding sides in the same order. $k > 1$ is an enlargement and $0 < k < 1$ is a reduction (P to S has $k = 1/2$). A single constant k for all lengths keeps the shape the same.	multiplying, not by adding, and the shape stays the same. Our first model explains lengths but not yet why Mr Otieno's price went from KES 30 to KES 120, so we need to investigate area next.
Lesson 5 Using LSF to Find Unknown Lengths: Photo L and the Kericho Tree Shadow	Photo L is 9 cm x 12 cm with diagonal 15 cm when S (3 x 4, diagonal 5) is enlarged with LSF 3. A 1.5 m student with a 2 m shadow and a tree with a 12 m shadow give $k = 12/2 = 6$.	If figures are similar, we find k from one pair of corresponding sides (image/object), then multiply object lengths by k to get image lengths, or divide image lengths by k to get object lengths. The tree height is $1.5 \times 6 = 9$ m.	Shadows work because the sun's rays make equal angles, so the student and tree triangles are similar. Every length in the photo follows the same LSF, so we can predict any missing side of a resized passport photo. We have not yet explained why the cost of P is KES 120 rather than double KES 30.
Lesson 6 Area Scale Factor and the Cost Puzzle: Why Does Doubling the Photo Quadruple the Price?	We drew prints S (3 x 4 cm), P (6 x 8 cm), L (9 x 12 cm) and H (1.5 x 2 cm) on grid paper and found areas of 12, 48, 108 and 3 cm ² . Four copies of S fit exactly on P, and nine copies fit on L. The prices at KES 2.50 per cm ² were KES 30, 120, 270 and 7.50. The distorted 6 x 4 cm photo had area 24 cm ² .	When a figure is enlarged with linear scale factor k, its area is multiplied by the area scale factor ASF = k^2 . For $k = 2$, ASF = 4; for $k = 3$, ASF = 9; for $k = 1/2$, ASF = $1/4$. Cost follows area, not length. The rule needs both dimensions to be multiplied by the same k (similar figures).	Achieng's print P has every length doubled (LSF 2), so its area is $2^2 = 4$ times that of S (12 cm ² to 48 cm ²). Mr Otieno charges KES 2.50 per cm ² , so the price rises from KES 30 to KES 120, which is four times and not double. The distorted photo has only the width doubled, so its area (24 cm ²) is only twice and it is not similar.
Lesson 7 Volume Scale Factor: From Photo Area to Cube and Jerrycan Volume	We built a 2 cm cube (8 cm ³) and a 4 cm cube (64 cm ³), compared the 5 L jerrycan with the similar jerrycan that holds 40 L, and calculated an LSF 3 cube of 216 cm ³ (27 times).	When every length is multiplied by the linear scale factor k, the volume is multiplied by VSF = k^3 . The pattern is length x k, area x k^2 and volume x k^3 , and it works only for similar objects.	Doubling every length of a solid gives $2 \times 2 \times 2 = 8$ times the volume, because length, width and height each double. That is why a similar jerrycan with LSF 2 holds 40 L, not 10 L or 20 L, and why Achieng's photo (LSF 2) costs 4 times as much because of ASF = 4. This adds volume to the model of her passport photo.
Lesson 8 Enlargement with a Positive Scale Factor: Locating the Centre and Drawing Rays	We drew rays from the centre O(0,0) through A(1,1), B(3,1) and C(2,3) and marked each image point at twice its distance from O. We got A'(2,2), B'(6,2), C'(4,6). Each side of the image was 2 times as long as the matching side of the object and the angles were equal.	An enlargement needs a centre and a scale factor. With a positive scale factor k, each image point lies on the ray from the centre through the object point, at k times the distance from the centre. The image is similar to the object, just like print P is similar to print S.	Achieng's print P is an enlargement of print S with scale factor 2. Every point of the picture is moved along a ray from the centre to twice the distance, so every length doubles and the angles stay the same, so the shape is unchanged. Doubling only the width (6 x 4 cm) does not follow this rule, so the photo is stretched and not similar.
Lesson 9 Shrinking and Flipping: Fractional and Negative Scale Factors	We drew rays from the centre O(0,0) through A(1,1), B(3,1), C(2,3). With $k = 2$ we got A'(2,2), B'(6,2), C'(4,6). With $k = 1/2$ applied to A'B'C' we returned to A(1,1), B(3,1), C(2,3). With $k = -2$ we got A''(-2,-2), B''(-6,-2), C''(-4,-6) on the opposite side of	If $0 < k < 1$, the image is a smaller copy on the same side of the centre (a reduction, still an enlargement with a fractional scale factor). If k is negative, the image is on the opposite side of the centre and inverted, and its size is given by $ k $. In both cases $k =$	Mr Otieno can shrink print P to print S using $k = 1/2$, so every length halves (6 to 3, 8 to 4, diagonal 10 to 5) while the angles stay the same, so the photo is not stretched. Distances from the centre are multiplied by k, which is why negative k flips the image to

	O, upside down. Print P (6 x 8 cm) to print S (3 x 4 cm) has $k = 1/2$.	image length / object length and the shape stays similar.	the opposite side of the centre.
Lesson 10 Finding the Centre and Scale Factor of an Enlargement from Object and Image	Lines drawn through corresponding points A and A', B and B', and C and C' all meet at one point, (1,0). Object AB = 2 and image A'B' = 4. Object and image have equal angles and parallel sides.	The point where lines through corresponding points meet is the centre of enlargement. The scale factor is $k = \text{image length} / \text{object length} = 4/2 = 2$. An enlargement is described by its centre and its scale factor.	Print P is an enlargement of print S with $k = 2$ (3 to 6, 4 to 8, 5 to 10), and the shape stays the same because every point moves along a line through the centre and every length is multiplied by the same factor. We can recover the centre and k from only the object and image.
Lesson 11 Maps, Scale Drawings and Costs: Using Scale 1:n in Nairobi, Mombasa and Kisumu	We measured 8.8 cm for Nairobi to Mombasa and 7 cm for Kisumu to Nairobi on a 1:5,000,000 map, and we found that the 1:50 classroom plan is 16 cm x 12 cm (192 cm ²) while the real room is 8 m x 6 m (48 m ²).	A map scale 1:n is a linear scale factor: real length = map length x n, so 8.8 cm gives 440 km and 7 cm gives 350 km. Area changes by n^2 ($50^2 = 2500$, so 192 cm ² becomes 480,000 cm ² = 48 m ²), and cost follows area (48 x KES 800 = KES 38,400).	This is the same rule as Achieng's passport photos: when the length is doubled the area is 4 times, so print P costs KES 120 and not KES 60. Maps, plans and photos are similar figures, lengths change by k, areas by k^2 , and cost depends on area.
Lesson 12 Consolidation: Explaining the Passport Photo	We observed that print P (6 x 8 cm) is an enlargement of print S (3 x 4 cm): every length is doubled (3 to 6, 4 to 8, diagonal 5 to 10 cm) and the angles stay the same. The area went from 12 cm ² to 48 cm ² and the price from KES 30 to KES 120. Print L (9 x 12 cm) has area 108 cm ² and costs KES 270. The 6 x 4 cm photo, where only the width was doubled, looked stretched. A similar jerrycan with LSF 2 holds 40 L instead of 5 L.	We learned that an enlargement keeps the same shape when corresponding angles are equal and corresponding sides are in the same ratio. LSF $k = \text{image length} / \text{object length}$. ASF = k^2 and VSF = k^3 . Cost at KES 2.50 per cm ² follows area, so it changes by k^2 , not k. We also learned that a negative or fractional scale factor still gives an enlargement, and that a centre and a scale factor can be found from an object and its image.	Achieng's photo can be made bigger or smaller without changing its shape by multiplying every length by the same linear scale factor. From S to P, $k = 2$, so lengths double, but area is multiplied by $2^2 = 4$ (12 cm ² to 48 cm ²). Mr Otieno charges KES 2.50 per cm ² , so the price rises from KES 30 to KES 120, which is four times and not double. For S to L, $k = 3$, ASF = 9 and the price is KES 270. If only the width is doubled (6 x 4 cm), the sides are not in the same ratio, so the photo is not similar. For solids, volume is multiplied by k^3 , so the jerrycan goes from 5 L to 40 L.

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